

Non-abelian bosonization from factored coset models in path integrals

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Abstract

We present a derivation of abelian and non-abelian bosonization in a path integral setting by expressing the generating functional for current-current correlation functions as a product of a G/G -coset model, which is dynamically trivial, and a bosonic part which contains the dynamics. A BRST symmetry can be identified which leads to smooth bosonization in both the abelian and non-abelian cases.

1. Introduction

Bosonization has proven to be one of the most important aspects of quantum field theory in two dimensions and after the work of Luther and Peschel [1], Coleman [2] and Mandelstam [3] the formalism was expanded and refined. These results are typically summarized in what has become known as a bosonization dictionary which captures the one-to-one relations between elements of the fermionic and bosonic descriptions and allows one to translate a fermionic theory into a bosonic theory. The dictionary equates for example the free massless Dirac field with a free massless bosonic field and it gives rules for how some fermionic operators can be replaced by bosonic ones. At the same time it provides a powerful tool to analyze fermionic models in two dimensions. A non-abelian extension of this approach has been given by Witten [4].

The versatility of path integrals clearly indicate the desirability to develop these techniques also in a path integral language. A path integral version of the bosonization technique for interacting models (like the Thirring and Schwinger models) had in fact been developed some time ago [5], but it was only recently [6] that (in the abelian

case) the equivalence between free bosons and free fermions was established with path integral techniques.

The central idea in Ref. [6] was to introduce bosonic degrees of freedom into the path integral in such a way that a gauge symmetry appears, so that the fermionic degrees of freedom can then be “frozen” out by fixing a gauge. Although conceptually appealing, this idea seems somehow difficult to implement in practice.

The aim of the present paper is to derive a bosonization dictionary from path integrals in an alternative way. Fermionic degrees of freedom are traded for bosonic degrees of freedom by rewriting the generating functional of the fermions in such a way that a fermionic G/G coset model [7] is factored out. This is achieved by first introducing a suitable identity followed by a chiral change of variables.

This method offers a number of advantages over the method of Ref. [6]. Firstly, it is technically much simpler, mainly because no gauge fixing is necessary and therefore the complications due to gauge fixing in [6] are avoided. More importantly, our method generalizes readily to the non-abelian case. Indeed, for fermions in a representation of a non-abelian group G , we are able to show the equivalence between a theory of free fermions and a G/G coset model times a Wess-Zumino-Witten (WZW) model [4]. In this way we rederive in a rather straightforward way Witten’s non-abelian bosonization [4] for free fermions in the path integral setting. Furthermore our method enables us to derive a complete bosonization dictionary in path integral language, that is, we also obtain the boson equivalent of fermionic bilinear products such as mass terms. To achieve this we specifically avoid integrating out the fermions and keep the fermionic degrees of freedom as part of a dynamically trivial G/G -coset model with precisely known contents.

Concerning the G/G coset model [8] which factors out in our approach, we note that it corresponds to a topological model, so that the associated energy momentum tensor is zero – no dynamical excitations exist in this sector. The original fermionic model is thus factored into a dynamically trivial model and a bosonic model which carries the physical degrees of freedom of the original fermions.

It should be noted that we concentrate on correlation functions of currents and other fermionic bilinear products. For simplicity we do not consider energy-momentum correlation functions, but our method can be extended to study such correlation functions in a straightforward way by starting with free fermions in a background metric. Here we employ a flat Euclidian space-time throughout.

Although our method is logically independent from the method in Ref. [6], there are interesting parallels. In particular, although we do not introduce a bosonization gauge symmetry from the start as in Ref. [6], we discover the existence of a BRST-symmetry which appears naturally in our work. This BRST symmetry is especially important in the non-abelian case where it guides us in handling a determinant.

The outline of the paper is as follows. In Section 2 we discuss our bosonization technique in detail in the abelian case. In Section 3 we discuss the BRST symmetry and some parallels with the work in Ref. [6]. The general formalism for the non-abelian case is presented in Section 4 and in Section 5 we summarize and draw some conclusions.

2. Abelian bosonization

For simplicity we first illustrate the method in the abelian case. We consider the problem of free Dirac fermions in two dimensional Euclidian space for which the generating functional for current-current correlation functions is

$$Z[A_\mu] = \int D\bar{\psi} D\psi \exp \left(- \int d^2x [\bar{\psi} i \gamma_\mu \partial_\mu \psi - j_\mu A_\mu] \right). \quad (1)$$

In holomorphic and anti-holomorphic coordinates this reads

$$\begin{aligned} Z[A_+, A_-] &= \int D\bar{\psi} D\psi \exp \left(- \int d^2x [\psi_-^\dagger i \partial_+ \psi_- + \psi_+^\dagger i \partial_- \psi_+ - A_- j_+ - A_+ j_-] \right), \end{aligned} \quad (2)$$

but it is general enough to consider

$$Z[A_+] = \int D\bar{\psi} D\psi \exp \left(- \int d^2x [\psi_-^\dagger i \partial_+ \psi_- + \psi_+^\dagger i \partial_- \psi_+ - A_+ j_-] \right). \quad (3)$$

This becomes transparent once we make a change of variables in Eq. (2), namely

$$\begin{aligned} \psi' &= \exp[i\eta(x)] \psi, \\ \bar{\psi}' &= \bar{\psi} \exp[-i\eta(x)], \end{aligned} \quad (4)$$

where $\eta(x)$ is chosen so that $\partial_- \eta(x) = -A_-$ and where we introduce $A'_+ = A_+ + \partial_+ \eta$. The fermionic measure is invariant under this transformation as it is a gauge transformation and Eq. (2) becomes

$$Z[A'_+] = \int D\bar{\psi}' D\psi' \exp \left(- \int d^2x [\psi_-'^\dagger i \partial_+ \psi'_- + \psi_+'^\dagger i \partial_- \psi'_+ - A'_+ j_-] \right). \quad (5)$$

Dropping primes for notational simplicity, this is just expression (3) as asserted. For the sake of technical convenience we use expression (3) rather than (2) and at the end of the section re-introduce the original sources and discuss how to bosonize without making transformation (4).

The key idea to our approach is to rewrite the generating functional (5) in the form

$$\int D\bar{\psi} D\psi DB_+ \exp(-S_{\text{cf}}) \times \int D\phi \exp(-S_{\text{B}}[\phi, A_+]), \quad (6)$$

where S_{cf} is the action of a constrained fermionic $U(1)/U(1)$ coset model which in light cone gauge reads (see Refs. [7,9])

$$S_{\text{cf}} = \int d^2x [\psi_-^\dagger i \partial_+ \psi_- + \psi_+^\dagger i \partial_- \psi_+ - B_+ j_- + \text{ghosts}], \quad (7)$$

with B_+ in the role of a Lagrange multiplier that enforces the coset condition on physical states [8,9]. S_{B} is a bosonic action to be determined and the source term A_+ is coupled only to the bosons of this model.

To achieve this we have to introduce a bosonic field into the generating functional (5) which will play the role of the field B_+ in (7). With this in mind, we insert the identity

$$1 = \int DB_+ \exp \left(\int d^2x B_+ j_- \right) \delta(B_+) \quad (8)$$

into the generating functional (5) which now becomes

$$Z[A_+] = \int D\bar{\psi} D\psi DB_+ \delta(B_+) \exp \left(- \int d^2x [L_F - A_+ j_- - B_+ j_-] \right). \quad (9)$$

where we have dropped the primes and introduced

$$L_F = \psi_-^\dagger i \partial_+ \psi_- + \psi_+^\dagger i \partial_- \psi_+. \quad (10)$$

We represent the delta functional as

$$\delta(B_+) = \int D\lambda_- \exp \left(\int d^2x B_+ \lambda_- \right) \quad (11)$$

and shift the field B_+ to $B'_+ = B_+ + A_+$, so that Eq. (9) becomes

$$Z[A_+] = \int D\bar{\psi} D\psi DB'_+ D\lambda_- \exp \left(- \int d^2x [L_F - B'_+ j_- - B'_+ \lambda_- + A_+ \lambda_-] \right). \quad (12)$$

Comparing $Z[A_+]$ above with Eqs. (6) and (7), we see that the first two terms of the action are in fact those of a constrained fermionic model. In order to cast Eq. (12) completely in the form of Eq. (7), the third term in Eq. (12) will have to vanish. This can be achieved by a chiral change of variables in Eq. (12), since a term linear in B'_+ is then generated as a result of the chiral anomaly which originates from the non-trivial way the fermionic measure transforms under a chiral rotation [10]. To see this, consider first an infinitesimal transformation

$$\begin{aligned} \psi'_- &= \exp[i\theta(x)] \psi_-, & \psi'_+ &= \psi_+ \\ \psi'^\dagger_- &= \psi_-^\dagger \exp[-i\theta(x)], & \psi'^\dagger_+ &= \psi_+^\dagger, \end{aligned} \quad (13)$$

which yields the Fujikawa Jacobian [10]

$$J_F = \exp \left(\frac{i}{2\pi} \int d^2x \theta \tilde{F} \right), \quad (14)$$

where

$$\begin{aligned} \tilde{F} &= \epsilon_{\mu\nu} \partial_\mu B'_\nu \\ &= -2i \partial_- B'_+. \end{aligned} \quad (15)$$

B'_μ is the field which couples to the current j_μ and appears in the covariant derivative used to regularize the Fujikawa Jacobian.

Extension to a finite chiral rotation yields the Jacobian

$$J_F = \exp \left(\frac{1}{\pi} \int d^2x \left(\frac{1}{2} \theta \partial_- \partial_+ \theta + \theta \partial_- B'_+ \right) \right). \quad (16)$$

This result is more complicated than the corresponding expression (14) for an infinitesimal transformation, because under a finite transformation the field B'_μ , which is coupled to j_μ and is used as a regulator, changes. Before exploiting expression (16) to address the unwanted linear term in expression (12), we present a short derivation of J_F . Introduce a chiral transformation

$$\begin{aligned} \psi_- &\rightarrow \exp[i\alpha\theta(x)] \psi_-, \\ \psi_-^\dagger &\rightarrow \psi_-^\dagger \exp[-i\alpha\theta(x)], \end{aligned} \quad (17)$$

with α a real number between 0 and 1, to obtain

$$\begin{aligned} \psi_-^\dagger i\partial_+ \psi_- + \psi_+^\dagger i\partial_- \psi_+ - B'_+ j_- \\ \rightarrow \psi_-^\dagger i\partial_+ \psi_- + \psi_+^\dagger i\partial_- \psi_+ - (B'_+ + \alpha\partial_+ \theta) j_-. \end{aligned} \quad (18)$$

For the associated infinitesimal transformation

$$\begin{aligned} \psi_- &\rightarrow \exp[i\Delta\alpha\theta(x)] \psi_-, \\ \psi_-^\dagger &\rightarrow \psi_-^\dagger \exp[-i\Delta\alpha\theta(x)], \end{aligned} \quad (19)$$

we correspondingly substitute $B'_+ \rightarrow B'_+ + \alpha\partial_+ \theta$ in Eq. (15) to obtain the contribution

$$J_F = \exp \left(\frac{1}{\pi} \int d^2x (\Delta\alpha \partial_- (B'_+ + \alpha\partial_+ \theta)) \right). \quad (20)$$

Summing over $\Delta\alpha$ now gives the contribution for the finite transformation,

$$\begin{aligned} J_F &= \exp \left(\frac{1}{\pi} \int d^2x \int_0^1 d\alpha (\alpha\theta \partial_- \partial_+ \theta + \theta \partial_- B'_+) \right) \\ &= \exp \left(\frac{1}{\pi} \int d^2x \left(\frac{1}{2} \theta \partial_- \partial_+ \theta + \theta \partial_- B'_+ \right) \right), \end{aligned} \quad (21)$$

as asserted.

Returning to Eq. (12) we see that a finite transformation (13) for θ in Eq. (12) yields

$$\begin{aligned} Z[A_+] &= \int D\bar{\psi}' D\psi' DB'_+ D\lambda_- \exp(-S'), \\ S' &= \int d^2x \left(\psi_-^{\dagger'} i\partial_+ \psi'_- + \psi_+^{\dagger'} i\partial_- \psi'_+ - (B'_+ + \partial_+ \theta) j'_- \right. \\ &\quad \left. + \frac{1}{2\pi} \partial_- \theta \partial_+ \theta + \frac{1}{\pi} B'_+ \partial_- \theta - B'_+ \lambda_- + A_+ \lambda_- \right). \end{aligned} \quad (22)$$

We are now almost in a position to show the equivalence of expressions (12) and (6). By making the change of variables

$$\lambda_- = \frac{1}{\pi} \partial_- \theta, \quad (23)$$

the second and third last terms cancel and we obtain

$$Z[A_+] = \int D\bar{\psi} D\psi DB'_+ D\theta \det\left(\frac{1}{\pi} \partial_-\right) \exp\left(-\int d^2x L\right) \quad (24)$$

with

$$L = L_F - (B'_+ + \partial_+ \theta) j_- + \frac{1}{2\pi} \partial_- \theta \partial_+ \theta + \frac{1}{\pi} A_+ \partial_- \theta, \quad (25)$$

while the determinant originating from the change of variables (23) can be represented by ghosts in the form $\int D\bar{c} Dc \exp(-\int d^2x \bar{c} \partial_- c)$.

Finally, shifting $B'_+ \rightarrow B'_+ - \partial_+ \theta$, we see that we have succeeded in our original aim: the free fermion generating functional has been written as the product of partition functions for a coset model and a theory of free bosons.

A point to note is that the coset model obtained above is in the light cone gauge. Normally the $U(1)/U(1)$ coset model is formulated in terms of fermions as [7,9]

$$L_{U(1)/U(1)} = \psi_-^\dagger \partial_+ \psi_- + \psi_+^\dagger \partial_- \psi_+ + B_- j_+ + B_+ j_- . \quad (26)$$

This model has a $U(1)$ gauge invariance which needs to be gauge fixed. If we choose the light cone gauge we have

$$\begin{aligned} Z_{U(1)/U(1)} &= \int D\bar{\psi} D\psi DB_+ DB_- D\bar{c} Dc \delta(B_-) \Delta_{\text{FP}} \exp\left(-\int d^2x L_{U(1)/U(1)}\right) \\ &= \int D\bar{\psi} D\psi DB_+ \exp\left(-\int d^2x [L_{U(1)/U(1)} + B_+ j_- + \bar{c} \partial_- c]\right), \end{aligned} \quad (27)$$

where Δ_{FP} is the Fadeev-Popov determinant. In the second line we have represented this determinant with ghosts. We note that the fermionic parts of expressions (24) and (27) are the same. The ghosts introduced to lift the Jacobian in expression (23) may be identified with the Fadeev-Popov ghosts in expression (27).

We now discuss how the product representation (24) of the fermionic generating functional can be used to extract the bosonization dictionary. To calculate current-current correlation functions, we only need to consider the bosonic part of Eq. (24), because the source is decoupled from the fermions in

$$Z[A'_+] = N \int D\theta \exp\left(-\int d^2x \left[+\frac{1}{2\pi} \partial_- \theta \partial_+ \theta + \frac{1}{\pi} A_+ \partial_- \theta\right]\right). \quad (28)$$

We have included in the normalization N the fermionic coset factor which, as stressed above, is irrelevant for current-current correlation functions. In order to rewrite our result in terms of the original sources of the generating functional (2), we substitute A_+ in the

above equation by $A_+ + \partial_+ \eta$ where $\partial_- \eta(x) = -A_-$ and perform a partial integration. This gives

$$Z[A_+, A_-] = N \int D\theta \exp \left(- \int d^2x \left[\frac{1}{2\pi} \partial_- \theta \partial_+ \theta + \frac{1}{\pi} A_\mu \epsilon_{\mu\nu} \partial_\nu \theta \right] \right), \quad (29)$$

so that we can associate the fermionic current with $\epsilon_{\mu\nu} \partial_\nu \theta$. In order to write the bosonization rule for the fermionic current with the usual normalization, we rescale the bosonic field according to $\theta/\sqrt{\pi} \rightarrow \theta$, so that we find from Eq. (29) the well known bosonization rule

$$j_\mu \rightarrow (1/\sqrt{\pi}) \epsilon_{\mu\nu} \partial_\nu \theta. \quad (30)$$

We have thus been able to rewrite the partition function for free fermions in the presence of an external source as the partition function for free bosons (times a coset model factor which plays no physical role). By just considering the coupling of the bosonic field to the source we could then derive the bosonization rule (30). Although our procedure may seem complicated for the simple abelian case, we shall see below that it is easily extended to the non-abelian case for the derivation of more involved non-abelian bosonization rules.

Concerning the bosonization rules for ψ , we just add source terms of the form $\psi_-^\dagger \psi_+$ and $\psi_+^\dagger \psi_-$ to the original fermion generating functional, extending it to

$$Z[A_+, \rho, \tilde{\rho}] = \int D\bar{\psi} D\psi \exp \left(- \int d^2x [L_F - A_+ j_- + \rho \psi_-^\dagger \psi_+ + \tilde{\rho} \psi_+^\dagger \psi_-] \right). \quad (31)$$

We then go through the same steps as above to obtain Eq. (24) with additional source terms

$$\begin{aligned} Z[A_+, \rho, \tilde{\rho}] \\ = \int D\bar{\psi} D\psi DB'_+ D\theta \exp \left(- \int d^2x [L_{CF} + L_B + \rho \exp(-i\theta) \psi_-^\dagger \psi_+ \right. \\ \left. + \tilde{\rho} \exp(i\theta) \psi_+^\dagger \psi_-] \right). \end{aligned} \quad (32)$$

If we now differentiate with respect to the sources and set them equal to zero, we again obtain a factorization into a coset model and a free bosonic model. As an example we calculate

$$\begin{aligned} \langle \psi_-^\dagger \psi_+(x_1) \psi_+^\dagger \psi_-(x_2) \rangle_{\text{free fermion}} \\ = \langle \psi_-^\dagger \psi_+(x_1) \psi_+^\dagger \psi_-(x_2) \rangle_{\text{coset}} \langle \exp[-i\theta(x_1)] \exp[i\theta(x_2)] \rangle_{\text{free boson}}. \end{aligned} \quad (33)$$

The correlator $\langle \rangle_{\text{coset}}$ is in fact a constant [9] because, as already stated, the coset model has no dynamical degrees of freedom. Note that we have to restrict our discussion here to gauge invariant fermionic quantities, as the correlators of quantities like ψ are

not constant in the coset model, but have a value which depends on how the gauge in the coset model has been fixed.

From Eq. (33) one can infer the bosonization rule

$$\psi_+^\dagger \psi_- \rightarrow \exp(i\theta). \quad (34)$$

At the canonical level ψ_+ is only a function of x_+ and $\psi_-^\dagger$ of x_- , which means that we may associate ψ_+ with the holomorphic part of $\exp(i\theta)$.

The fact that we have obtained, as a matter of convenience, a coset model in a particular gauge, may raise questions about the generality of the result. It is in fact possible to modify the procedure to obtain a coset model where the gauge has not been fixed. This is achieved by noting that in

$$Z[A_\mu] = \int D\bar{\psi} D\psi \exp\left(-\int d^2x [\bar{\psi}\gamma_\mu i\partial_\mu\psi - j_\mu A_\mu]\right), \quad (35)$$

we can make the following change of variables (which corresponds to a gauge transformation)

$$\begin{aligned} \psi &\rightarrow \exp[i\eta(x)] \psi, \\ \bar{\psi} &\rightarrow \bar{\psi} \exp[-i\eta(x)]. \end{aligned} \quad (36)$$

to obtain

$$Z[A_\mu] = \int D\bar{\psi} D\psi \exp\left(-\int d^2x [\bar{\psi}i\gamma_\mu\partial_\mu\psi - j_\mu A_\mu - \partial_\mu\eta j_\mu]\right). \quad (37)$$

Since this generating functional is in fact η -independent, we find after integrating over η (and absorbing an infinite normalization constant)

$$Z[A_\mu] = \int D\bar{\psi} D\psi \delta(\partial_\mu j_\mu) \exp\left(-\int d^2x [\bar{\psi}i\gamma_\mu\partial_\mu\psi - j_\mu A_\mu]\right). \quad (38)$$

We now substitute the delta functional

$$\delta(\partial_\mu j_\mu) = \int DB_\mu D\theta \exp\left(\int d^2x \left[B_\mu j_\mu + \frac{1}{\pi} B_\mu \epsilon_{\mu\nu} \partial_\nu \theta\right]\right) \quad (39)$$

in Eq. (38) to obtain

$$\begin{aligned} Z[A_\mu] = \int D\bar{\psi} D\psi DB_\mu D\theta \exp\left(-\int d^2x \left[\bar{\psi}i\gamma_\mu\partial_\mu\psi - j_\mu(A_\mu + B_\mu) \right. \right. \\ \left. \left. - \frac{1}{\pi} B_\mu \epsilon_{\mu\nu} \partial_\nu \theta\right]\right). \end{aligned} \quad (40)$$

(The identity (39) can be obtained by first integrating over the field θ and then over B_μ .)

We can now obtain a constrained fermionic model by introducing the change of variables

$$\begin{aligned}
\psi &\rightarrow \exp[i\gamma_5\theta'(x)] \psi, \\
\bar{\psi} &\rightarrow \bar{\psi} \exp[-i\gamma_5\theta'(x)] \\
\theta(x) &\rightarrow \theta'(x),
\end{aligned} \tag{41}$$

The last term in Eq. (40) then cancels the linear term in B of the Fujikawa determinant so that we finally have

$$Z[A_\mu] = \int D\bar{\psi} D\psi DB_\mu D\theta \exp\left(-\int d^2x L\right) \tag{42}$$

with

$$L = \bar{\psi} i\gamma_\mu \partial_\mu \psi - j_\mu (A_\mu + B_\mu + \epsilon_{\mu\nu} \partial_\nu \theta) + \frac{1}{\pi} A_\mu \epsilon_{\mu\nu} \partial_\nu \theta + \frac{1}{2\pi} \partial_\nu \theta \partial_\nu \theta, \tag{43}$$

where we have used $\gamma_\mu \gamma_5 = \epsilon_{\mu\nu} \gamma_\nu$ and partial integrations.

Eq. (42) is the desired result. Shifting the field $B_\mu \rightarrow A_\mu + B_\mu + \epsilon_{\mu\nu} \partial_\nu \theta$ we obtain a constrained fermionic model which is not gauge fixed together with a free bosonic theory. The procedure is simpler, however, if we work only with right-handed fermions at intermediate stages, especially in the non-abelian case, as becomes apparent in Section 4.

3. BRST symmetry and smooth bosonization

In this section we briefly discuss the relation between our procedure and the smooth bosonization approach of Ref. [6], where a gauge symmetry, referred to as “bosonization gauge symmetry”, was introduced by making a γ_5 transformation and integrating over the corresponding angle. This procedure indicates that bosonization can be viewed as arising from the choice of a particular gauge. Although no gauge symmetry is introduced in our approach, a similar view of bosonization can be taken by noting that a BRST symmetry arises naturally within our scheme.

Consider again Eq. (12)

$$Z[A_+] = \int D\bar{\psi} D\psi DB_+ D\lambda_- \exp\left(-\int d^2x [L_F - A_+ j_- - B_+ j_- - B_+ \lambda_-]\right). \tag{44}$$

Making the change of variables

$$\begin{aligned}
\psi'_- &= \exp[i\theta(x)] \psi_-, & \psi'_+ &= \psi_+, \\
\psi'^{\dagger}_- &= \psi^{\dagger}_- \exp[-i\theta(x)], & \psi'^{\dagger}_+ &= \psi^{\dagger}_+, \\
\lambda_- &= \lambda_-(\theta),
\end{aligned} \tag{45}$$

we obtain

$$Z[A_+] = \int D\bar{\psi} D\psi DB_+ D\theta D\bar{c} Dc \exp(-S),$$

$$S = \int d^2x \left[L_F - (B_+ + A_+ + \partial_+ \theta) j_- + \frac{1}{2\pi} \partial_- \theta \partial_+ \theta \right. \\ \left. + \frac{1}{\pi} (B_+ + A_+) \partial_- \theta - B_+ \lambda_- (\theta) + \bar{c} \frac{\delta \lambda_- (\theta)}{\delta \theta} c \right], \quad (46)$$

where we have lifted the Jacobian induced by $\lambda_- \rightarrow \lambda_-(\theta)$ with ghosts. Apart from the last two terms the above generating functional has a gauge invariance under

$$\begin{aligned} \psi_- &\rightarrow \exp[i\alpha(x)] \psi_-, & \psi_+ &\rightarrow \psi_+, \\ \psi_-^\dagger &\rightarrow \psi_-^\dagger \exp[-i\alpha(x)], & \psi_+^\dagger &\rightarrow \psi_+^\dagger, \\ \theta(x) &\rightarrow \theta(x) - \alpha(x). \end{aligned} \quad (47)$$

This can easily be verified explicitly: the part of the Lagrangian that contains fermions is exactly invariant under the transformation while the variation of the bosonic part cancels the contribution coming from the fermionic measure. Only the last two terms break the symmetry explicitly. The existence of this symmetry stems from the chiral transformation we made and the fact that we also integrate over the parameter of the transformation.

Although the total generating functional is not invariant under the transformation Eq. (47), a global BRST symmetry exists which leaves the generating functional invariant. This follows once we note that the last two terms may be written as a BRST exact form

$$\delta(\bar{c} \lambda_-) = B_+ \lambda_- + \bar{c} \frac{\delta \lambda_-}{\delta \theta} c, \quad (48)$$

where (with a global Grassmann number omitted for simplicity)

$$\begin{aligned} \delta \bar{c} &= B_+, & \delta \psi_- &= ic \psi_-, \\ \delta c &= 0, & \delta \psi_-^\dagger &= -i \psi_-^\dagger c, \\ \delta \theta &= -c, & \delta \psi_+ &= 0, \\ \delta B_+ &= 0, & \delta \psi_+^\dagger &= 0, \\ \delta A_+ &= 0. \end{aligned} \quad (49)$$

This BRST symmetry is nilpotent and therefore the variation of Eq. (48) is zero. The remainder of the partition function is invariant because this BRST symmetry is only a special case of the symmetry (47). The terms in expression (48) can therefore be viewed as a type of gauge fixing condition and a Fadeev-Popov determinant, respectively. Note however that no notion of anomalous gauge fixing as discussed in [6] is required in our approach.

If we now choose $\lambda_- = (1/\pi) \partial_- \theta$ in Eq. (46), we obtain the bosonization results of the previous section. We have the freedom, however, of choosing a still more general $\lambda_-(\theta)$. This will correspond to the smooth bosonization of [6]. As an example, instead of the identity (8) we can use

$$1 = \int DB_+ \exp \left(- \int d^2x \alpha B_+ j_- \right) \delta(B_+) , \quad (50)$$

where α is a parameter between 0 and 1. Eq.(46) becomes

$$Z[A_+] = \int D\bar{\psi} D\psi DB_+ D\theta D\bar{c} Dc \exp(-S) \quad (51)$$

with

$$S = \int d^2x \left[L_F - (\alpha B_+ + A_+ + \partial_+ \theta) j_- + \frac{1}{2\pi} \partial_- \theta \partial_+ \theta + \frac{1}{\pi} (\alpha B_+ + A_+) \partial_- \theta - B_+ \lambda_- + \bar{c} \frac{\delta \lambda_-}{\delta \theta} c \right]. \quad (52)$$

The choice $\lambda_- = (1/\pi) \partial_- \theta$ now leads for $\alpha = 1$ to a constrained fermion model and a free boson model, and in the case $\alpha = 0$ to a free fermion model and a trivial boson theory. For values of α between 0 and 1 one finds non-trivial models of fermions and bosons coupled together.

4. Non-abelian bosonization

In this section we generalize our approach to bosonization to the non-abelian case. One of the advantages of our scheme is in fact the simplicity with which abelian bosonization can be extended to the case of fermions in some representation of a non-abelian group G . We start with a theory of free Dirac fermions which are in the fundamental representation of $U(N)$. The generating functional is

$$Z[A_+, A_-] = \int D\bar{\psi} D\psi \exp \left(- \int d^2x [\psi_-^\dagger i \partial_+ \psi_- + \psi_+^\dagger i \partial_- \psi_+ - \psi_-^\dagger A_+ \psi_- - \psi_+^\dagger A_- \psi_+] \right), \quad (53)$$

where ψ is in the fundamental representation and $A = A^a T^a$.

As in the abelian case we find it convenient to set one of the source terms equal to zero at intermediate stages of the calculation. This can be accomplished by the following gauge transformation which we view as a change of variables

$$\begin{aligned} \psi'_- &= h \psi_- , & \psi'_+ &= h \psi_+ , \\ \psi'^{\dagger}_- &= \psi_-^\dagger h^{-1} , & \psi'^{\dagger}_+ &= \psi_+^\dagger h^{-1} . \end{aligned} \quad (54)$$

We choose h so that

$$h^{-1} A_- h - i h^{-1} \partial_- h = 0 \quad (55)$$

and introduce $A'_+ = h^{-1} A_+ h - i h^{-1} \partial_+ h$. The fermionic measure is invariant under this type of transformation. In the following we again drop the primes for notational simplicity.

To rewrite Eq. (53) as a functional which factorizes into a constrained fermionic model and a bosonic part, requires once again a field B_+ which will become the Lagrange multiplier of the constrained fermions. The identity (8) generalizes in an obvious way to

$$\begin{aligned} 1 &= \int DB_+ \delta(B_+) \exp \left(\int d^2x \psi_-^\dagger B_+ \psi_- \right) \\ &= \int DB_+ D\lambda_- \exp \left(\int d^2x [\psi_-^\dagger B_+ \psi_- + \text{Tr}(B_+ \lambda_-)] \right), \end{aligned} \quad (56)$$

where $B_+ = B_+^a T^a$ and $\lambda_- = \lambda_-^a T^a$.

This identity is introduced in Eq. (53) and to constrain the fermions we perform the chiral change of variables

$$\begin{aligned} \psi'_- &= g\psi_-, & \psi'_+ &= \psi_+, \\ \psi'^{\dagger}_- &= \psi_-^\dagger g^{-1}, & \psi'^{\dagger}_+ &= \psi_+^\dagger, \\ \lambda &= \lambda(g). \end{aligned} \quad (57)$$

The fermionic measure is not invariant under the transformation, since under an infinitesimal transformation

$$\begin{aligned} \psi_- &\rightarrow \exp[i\theta(x)] \psi_-, \\ \psi_-^\dagger &\rightarrow \psi_-^\dagger \exp[-i\theta(x)], \end{aligned} \quad (58)$$

where $\theta = \theta^a T^a$, one already infers the non-trivial Jacobian $J_F = (i/2\pi)\theta\tilde{F}$, with

$$\begin{aligned} \tilde{F} &= \epsilon_{\mu\nu} F_{\mu\nu}, \\ F_{\mu\nu} &= \partial_\mu A_\nu - \partial_\nu A_\mu + i[A_\mu, A_\nu]. \end{aligned} \quad (59)$$

Here A_μ is the field appearing in the covariant derivative used as a regulator when the Jacobian is calculated. In the present case $J_F = (1/\pi)\theta\partial_-(B_+ + A_+)$.

The corresponding determinant for a finite transformation may be calculated as in the abelian case. Here we exploit the result [11,5]

$$\det(i\partial_+ - A_+) = \exp(S[g]), \quad (60)$$

where g is defined by $A_+ = -ig^{-1}\partial_+g$, and $S[g]$ is the Wess-Zumino-Witten action

$$\begin{aligned} S[g] &= -\frac{1}{8\pi} \int d^2x \text{Tr}(g \partial_\mu g^{-1} g \partial_\mu g^{-1}) \\ &\quad - \frac{i}{12\pi} \int_\Gamma d^3x \epsilon_{\mu\nu\rho} \text{Tr}(g \partial_\mu g^{-1} g \partial_\nu g^{-1} g \partial_\rho g^{-1}). \end{aligned} \quad (61)$$

(Γ is a three dimensional ball with two dimensional spacetime as boundary.) This action satisfies the Polyakov-Wiegmann identity [11]

$$S[hg] = S[h] + S[g] - \frac{1}{4\pi} \int d^2x \text{Tr}[h^{-1}(\partial_+h)(\partial_-g)g^{-1}]. \quad (62)$$

The Jacobian J appearing in

$$\begin{aligned} & \int D\bar{\psi} D\psi \exp \left(- \int d^2x [\psi_-^\dagger i\partial_+ \psi_- + \psi_+^\dagger i\partial_- \psi_+ - \psi_-^\dagger (A_+ + B_+) \psi_-] \right) \\ &= \int D\bar{\psi} D\psi J \exp \left(- \int d^2x [\psi_-^\dagger g^{-1} i\partial_+ (g\psi_-) + \psi_+^\dagger i\partial_- \psi_+ \right. \\ & \quad \left. - \psi_-^\dagger g^{-1} (A_+ + B_+) g\psi_-] \right) \end{aligned} \quad (63)$$

can then be determined after integrating over the fermions, namely

$$\begin{aligned} J &= \frac{\det(i\partial_+ - (A_+ + B_+))}{\det(i\partial_+ - (g^{-1}A_+g + g^{-1}B_+g) + g^{-1}i\partial_+g)} \\ &= \frac{\exp(S[h])}{\exp(S[hg])} \\ &= \exp(S[h] - S[hg]) \\ &= \exp \left(-S[g] + \frac{i}{4\pi} \int d^2x \text{Tr}((A_+ + B_+)(\partial_-g)g^{-1}) \right), \end{aligned} \quad (64)$$

where $A_+ + B_+ = -ih^{-1}\partial_+h$.

From the change of variables (57) we therefore obtain

$$Z[\Lambda_+] = \int D\bar{\psi} D\psi DB_+ Dg \det \left(\frac{\delta\lambda_-}{\delta g} \right) \exp(-S) \quad (65)$$

with

$$\begin{aligned} S &= S_F + S[g] + \int d^2x \left[\psi_-^\dagger (g^{-1}B_+g + g^{-1}A_+g - ig^{-1}\partial_+g)\psi_- \right. \\ & \quad \left. - \frac{i}{4\pi} \text{Tr}((A_+ + B_+)(\partial_-g)g^{-1}) + \text{Tr}(B_+\lambda_-(g)) \right], \end{aligned} \quad (66)$$

where $\det(\delta\lambda_-/\delta g)$ is the Jacobian associated with the transformation from $D\lambda_-$ to the invariant measure Dg , while S_F is the free fermion action.

As in the abelian case this generating functional is “nearly” gauge invariant. Apart from the last term in the action and the determinant $\det(\delta\lambda_-/\delta g)$ we have invariance under the transformation

$$\begin{aligned} \psi_- &\rightarrow h\psi_-, & \psi_+ &\rightarrow \psi_+, \\ \psi_-^\dagger &\rightarrow \psi_-^\dagger h^{-1}, & \psi_+^\dagger &\rightarrow \psi_+^\dagger, \\ g &\rightarrow gh^{-1}. \end{aligned} \quad (67)$$

It is straightforward to check that the part of the Lagrangian that contains fermions is invariant, while the variation of the bosonic part precisely cancels the anomaly arising from the non-invariance of the fermionic measure. The total generating functional is,

however, not gauge invariant because of the term $\text{Tr}(B_+ \lambda_- (g))$ and the above mentioned determinant, which break the gauge symmetry explicitly.

A BRST invariance does, however, exist. As in the abelian case the part of the generating functional which is not gauge invariant can be written as a BRST exact form. To see this, note that the determinant $\det(\delta\lambda_-/\delta g)$ is, like the familiar Fadeev-Popov determinant, invariant under group transformations

$$D\lambda_- = \det\left(\frac{\delta\lambda_-}{\delta g}\right) Dg = \det\left(\frac{\delta\lambda_-}{\delta g'}\right) Dg', \quad (68)$$

because of the invariance of the Haar measure, and can therefore be evaluated close to the identity. We may parametrize g by $g = \exp(i\alpha^a T^a)$ and close to the identity the invariant measure is $D\alpha^a$. The determinant for the change of variables from λ_- to α is then $\det(\delta\lambda_-/\delta\alpha)$. This determinant is now lifted with ghosts. The complete generating functional is now invariant under the set of transformations

$$\begin{aligned} \delta\bar{c} &= B_+, & \delta\psi_- &= ic\psi_-, \\ \delta c &= ic^2, & \delta\psi_-^\dagger &= -i\psi_-^\dagger c, \\ \delta g &= -igc, & \delta\psi_+ &= 0, \\ \delta g^{-1} &= icg^{-1}, & \delta\psi_+^\dagger &= 0, \\ \delta B_+ &= 0, & \delta A_+ &= 0. \end{aligned} \quad (69)$$

In terms of the α this reads $\delta\alpha = -c$ and we have

$$\delta(\text{Tr}(\bar{c}\lambda_-)) = \text{Tr}\left(B_+ \lambda_- + \bar{c} \frac{\delta\lambda_-}{\delta\alpha} c\right), \quad (70)$$

which is exactly the gauge non-invariant part. Because the BRST symmetry is nilpotent the BRST variation of this term is zero. The remainder of the generating functional is invariant because the BRST transformations is a special case of the gauge transformation (67). The fermionic generating functional can therefore be written as

$$Z[A_+] = \int D\bar{\psi} D\psi DB_+ Dg D\bar{c} Dc \exp(-S), \quad (71)$$

with

$$\begin{aligned} S = S_F + S[g] + \int d^2x \left[\psi_-^\dagger (g^{-1} B_+ g + g^{-1} A_+ g - ig^{-1} \partial_+ g) \psi_- \right. \\ \left. - \frac{i}{4\pi} \text{Tr}((A_+ + B_+) (\partial_- g) g^{-1}) + \delta(\text{Tr}(\bar{c}\lambda_-)) \right]. \end{aligned} \quad (72)$$

For a general $\lambda_- = \lambda_-(g)$ we have a complicated interacting fermion and boson model, but we can factor out a constrained fermionic model if we choose

$$\lambda_-(g) = \frac{i}{4\pi} (\partial_- g) g^{-1} \quad (73)$$

to cancel the linear term $B_+(\partial_-g)g^{-1}$ in Eq. (72). For this choice of λ_- we have

$$\delta(\text{Tr}(\bar{c}\lambda_-)) = \text{Tr}(B_+\lambda_-) + \frac{1}{4\pi} \text{Tr}(g^{-1}\bar{c}g\partial_-c). \quad (74)$$

Making the change of variables $\bar{c}' = g^{-1}\bar{c}g$, a transformation with trivial determinant, the ghosts and bosons decouple and Eq. (66) simplifies to

$$Z[A_+] = \int D\bar{\psi} D\psi DB_+ Dg D\bar{c}' Dc \exp(-S) \quad (75)$$

with

$$S = S_F + S[g] + \int d^2x \left[\psi_-^\dagger (g^{-1}B_+g + g^{-1}A_+g - ig^{-1}\partial_+g)\psi_- - \frac{i}{4\pi} \text{Tr}((A_+)(\partial_-g)g^{-1}) + \bar{c}'\partial_-c \right]. \quad (76)$$

We now introduce a new Lagrange multiplier

$$B' = g^{-1}B_+g + g^{-1}A_+g - ig^{-1}\partial_+g, \quad (77)$$

a transformation for which the Jacobian is again trivial. We then obtain the final result

$$Z[A_+] = \int D\bar{\psi} D\psi DB_+ Dg D\bar{c}' Dc \exp(-S) \quad (78)$$

with

$$S = S_F + S[g] + \int d^2x \left[\psi_-^\dagger B'_+\psi_- + \bar{c}'\partial_-c - \frac{i}{4\pi} \text{Tr}(A_+(\partial_-g)g^{-1}) \right]. \quad (79)$$

In the literature [9] one finds the Jacobian $\det(i\partial_+ + ig^{-1}\partial_+g)$ associated with the change of variables from DA_+ to Dg (where $A_+ = -ig^{-1}\partial_+g$). This is formally equivalent to our result, but in the case of [9] the determinant is regularized to respect a gauge symmetry. In our case λ is not a gauge connection and there is no reason to regularize so as to respect gauge invariance. Instead we have a global BRST symmetry which should be maintained.

The interpretation of Eq. (78) is as follows. The fermionic part has decoupled from the bosonic part and it is a G/G coset model which is trivial from the dynamical point of view. All the dynamics now resides in the bosonic sector which corresponds to a WZW model at level one with the source of the fermionic current now coupled to the bosonic current $(\partial_-g)g^{-1}$. As in the abelian case, we immediately infer the bosonization rule

$$j_- \rightarrow i(\partial_-g)g^{-1} \quad (80)$$

for the j_- component of the fermionic current.

This model can be rewritten in terms of the original sources by making the vector gauge transformation hgh^{-1} and using the invariance of the Haar measure. For the bosonic part of the action we find

$$Z[A_+] = \int Dg \exp \left(-S[g] - \frac{i}{4\pi} \int d^2x \left[\text{Tr}(A_+(\partial_- g)g^{-1}) + \text{Tr}(A_-g^{-1}\partial_+ g) + A_+gA_-g^{-1} + A_+A_- \right] \right) \quad (81)$$

in terms of the original sources.

To calculate correlators of fermionic bilinears like $\psi_-^\dagger \psi_+$ and $\psi_+^\dagger \psi_-$, we simply add the corresponding source terms to Eq. (53). Under transformation (57) these terms transform to $\psi_+^\dagger g \psi_-$ and $\psi_-^\dagger g^{-1} \psi_+$. Differentiating with respect to the sources and then setting them equal to zero, we note that, as in Eq. (33), the correlators factorize into a fermionic part which is a constant, and correlators of g in the WZW model. Note that in the bosonic version of the coset model as a gauge WZW model, correlators of $\text{tr}(g)$ are constant [12], which corresponds to correlators of $\psi_+^\dagger \psi_-$ in the above fermionic coset.

5. Conclusions

In summary we have given within the path integral framework a complete derivation of the bosonization dictionary in 1+1 dimensions for both the abelian and non-abelian cases. This was achieved by inserting an appropriate identity into the generating functional for free fermions to introduce bosonic degrees of freedom. Then, after a chiral change of variables, the generating functional can be factorized in terms of a G/G -coset model partition function, which is dynamically trivial, and a bosonic model partition function.

We have illustrated our method with an external source A_μ . However, it is also applicable when A_μ is a background field or a dynamical field, that is, when we integrate over A_μ . This makes it possible to apply our method to interacting models like QCD_2 or the chiral Gross-Neveu model.

A further interesting aspect concerns the bosonization BRST symmetry discovered in Ref. [6]. This symmetry emerges quite naturally in our approach which, as we have shown, allows one to extend smooth bosonization to non-abelian systems in a straightforward manner.

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Note added

After completing this manuscript we became aware of Refs. [13–16] where similar results are obtained.

In Ref. [17,13] bosonization is viewed as a special case of a duality transformation. These authors “dualize” the free fermions by introducing a gauge field and constraining this gauge field to be a flat connection. They then perform the functional integration over the fermions and the gauge field to obtain a bosonized theory. In Ref. [16] a somewhat similar approach is taken.

The main difference between our work and this approach is that we specifically avoid integrating out the fermions, since it is more convenient to keep them as part of a trivial coset model when we want to obtain the bosonic image of some bilinear fermion operators, such as mass terms.

The identity (39) that we motivate and use in Section 2 to factor out the coset model that is not gauge fixed, effectively plays the role of the duality transformation of Refs. [17,13,15], in the sense that when this identity is inserted into the path integral (see Eq. (40)) the field θ may be considered to be a lagrange multiplier that enforces the constraint that the field B_μ is a flat connection. We find it however, more convenient not to work with this identity directly (especially in the non-Abelian case), but rather make a change of variables (54) first and then insert an identity in the form (56). Furthermore, to obtain the smooth bosonization results of Section 3, we require a more general identity than (39), namely (50). It is thus useful to insert the bosonic degrees of freedom into the path integral in a more general way than in Refs. [17,13] and from this point of view we cannot see any specific gain associated with the duality transformation.

We believe that it might also simplify matters in higher dimensions to make a change of variables of the form (54) before inserting an appropriate identity. Bosonization may be generalized to higher dimensions by inserting an identity like (39) in the path integral, integrating out the fermions formally [15] and treating the resulting determinant in some approximation scheme. In order to obtain some exact results in 1+1 dimensions in this article, we pursued, as already mentioned, a strategy which avoids integrating out the fermions. At this stage we do not yet have a similar procedure to bosonize by factoring out a dynamically trivial (topological) model in dimensions higher than two. (It is possible in 0+1 dimensions [18].)

In spirit our approach is closer to the one of Refs. [6,14]. In this formalism a gauge symmetry is introduced by hand and a gauge condition is then chosen so as to gauge fix the fermions out of the model. The fermions are therefore maintained as part of a gauge fixed model. The disadvantage of this formalism is, however, that in general the precise contents of this gauge fixed model is not known. It is important, though, to be able to analyse the contents of this model, because in the presence of fermionic sources the fermionic and bosonic sectors do not completely decouple and knowledge of the fermionic model is then necessary to evaluate some correlators (like those in Eq. (33)). In our approach the G/G -coset model which we factorize out is a well studied model [7,9,12] and its contents precisely known. This enables us to study correlators of fermionic bilinears and so derive the complete bosonization dictionary in the path integral language.

Naively our approach may seem to be more restrictive from the point of view that the procedure of Ref. [6] has a greater freedom in the sense of choosing different gauges. We can accommodate most of these, however, by just considering a slight variation in

the form of the identity we insert, as explained in Section 3. In this way we discover all the interesting features of [6,14], like smooth bosonization and the BRST symmetry, while avoiding the technical complications due to gauge fixing. Our work also provides a link between the work of [17,13] and [6,14] in the sense that we have shown how to derive the BRST symmetry and smooth bosonization of Ref. [6] using Eq. (39) which effectively implements the duality transformation of [17].

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